The mini-project report and source code must be submitted through the provided link.

https://forms.gle/z2rQUyy4YmZagxg78

Note: You may use **Arabic, French, or English** for writing the report.

PDF Report on the Work Must Include:

1. **A brief description of the core idea** on which the round design is based.
2. **The order of operations inside a round**, clearly explaining how the transformations are applied.
3. **S-Box and P-Box description**, including:
   - Their structure
   - The principle or rationale used to design them.
4. **Pseudocode for a single round** of the algorithm.
5. **The basis for key derivation**, explaining how round keys are generated from the main key.
6. **Simple pseudocode for the key derivation process.**
7. **Overall evaluation of the algorithm with screenshots**, using the **evaluation table provided in the guidelines**.
8. **Examples with screenshots** showing:
   - Plaintext
   - Corresponding cipher text.
9. **A short description of your development experience**, explaining the process, challenges encountered, and observations during the implementation.

Source code (with explanatory comments)
Allowed languages in order of preference:

1. Python
2. JavaScript
3. C#
4. C++
5. C
6. Java

Deadline : 01/04/2026

Evaluation

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Criterion	Description	Points
Implementation	The algorithm runs correctly and produces valid encryption results.	**2**
Code Structure	Clear design (functions/classes), logical organization of the algorithm.	**2**
Round Function	Correct implementation of the round operations (S-Box, P-Box, mixing, etc.).	**2**
Key Schedule	Proper implementation of the key derivation mechanism.	**2**
Testing & Examples	test examples demonstrating plaintext → cipher text.	**2**

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Criterion	Description	Points
Algorithm Concept	Clear explanation of the main idea behind the round design.	**2**
Round Structure	Detailed explanation of the order of operations in a round.	**2**
S-Box & P-Box Design	Clear description of their structure and the rationale behind their design.	**2**
Pseudocode	Correct pseudocode for the round function and key derivation.	**1**
Algorithm Evaluation	Proper use of the evaluation table provided in the guidelines.	**1**
Examples	Clear plaintext/cipher text examples with screenshots.	**1**
Development Experience	Reflection on implementation challenges and observations.	**1**